

NGUYEN GIA HUY

Java Backend Engineer

johnnynguyen882@gmail.com | 0948385149 | linkedin.com/in/huy8bit

PROFESSIONAL SUMMARY

Backend engineer moving into Java and Spring Boot, with nearly three years of production backend experience in Golang and Python on transactional, integration-heavy systems: lending and overdraft transaction logic on Thought Machine Vault Core at a digital bank, integrations between the core banking platform and external services, a microservices ecosystem of **100+ services** behind Kong Gateway, Keycloak-based multi-tenant authorization for **20+ organizations**, and a Kafka, Redis, and MongoDB backend serving **1.5K requests/second**. Working knowledge of **Java** and **Spring Boot**; the production systems below were built in Golang and Python.

TECHNICAL SKILLS

Languages:	Java, Spring Boot (working knowledge); Golang, Python, SQL (production)
APIs & Architecture:	REST/JSON, gRPC, microservices, API gateway, event-driven services
Security:	Authentication and authorization, Keycloak, RBAC, multi-tenancy
Data:	PostgreSQL, Amazon RDS, Redis, MongoDB, OpenSearch
Messaging:	Kafka, RabbitMQ
Testing & Delivery:	Automated simulation testing, CI/CD, Docker, Kubernetes, Jenkins, Git
Cloud:	AWS (EKS, ECS, RDS, Lambda, ECR, ELB)

WORK EXPERIENCE

Backend Engineer, Core Banking (Middle Software Engineer) 03/2026 – Present
Galaxy FinX (Vikki Digital Bank) | *core banking platform built on Thought Machine Vault Core*

Transaction Processing

- Modeled **financial transactions** for the Lending and Overdraft products on Thought Machine Vault Core, implemented and maintained as Python Smart Contracts.
- Implemented interest accrual and penalty and late-fee calculations, where correctness directly determines what customers owe.
- Implemented the **repayment and settlement** flows that close out each lending and overdraft obligation.

Integration and Data

- Integrated the core banking platform with **external backend services** that drive the bank's operational and transaction workflows.
- Cut account migration from **2.5 hours to 27 minutes (5x)** with a dedicated migration system over **Amazon RDS**, written in Golang and run on ECS and Lambda.

Testing and Delivery

- Rewrote the automated test harness for product simulations, moving it from Gauge to a **Golang** framework.
- Made simulation test runs **20x faster**, from 24 minutes to just over 50 seconds.
- Shipped a new Vault Core version to **Amazon EKS**, with container images in ECR, RDS as its database, and traffic through ELB.
- Built the **CI/CD pipeline** that automates build, test, and deployment.

Software Engineer – Backend & Data Services 04/2025 – 03/2026
Vietlink (KDDI Agile Development Center Group) | *au Denki – large-scale electric utility platform serving millions of users*

- Reworked **data access patterns** and parallelized batch processing as part of a pipeline redesign, increasing throughput by **150%**.
- Migrated the search service to **OpenSearch** from Elasticsearch as part of modernizing the analytics stack.
- Redesigned its data ingestion and indexing for consistency, stability, and query performance.
- Developed ETL services on AWS Lambda and ECS with RDS as the relational store and S3 with Athena for analytics, processing large-scale utility data.
- Operated the services in production with Terraform and AWS SAM deployments and CloudWatch and DataDog monitoring of health, failures, and anomalies.

Software Engineer – Microservices & Identity

04/2024 – 04/2025

VNPAY | *VNPAY Cloud – cloud platform for fintech and enterprise customers*

- Ran the **API gateway** (Kong) in front of a **microservices ecosystem of 100+ services**.
- Centralized service-level and network-level access control in the gateway layer.
- Implemented **authentication** for a multi-tenant platform on **Keycloak**, serving **20+ customer organizations**.
- Implemented the **authorization model** with role-based access control, keeping each organization isolated from the others.
- Operated **50+ production Kubernetes clusters** for VNPAY Cloud on OpenStack, with provisioning, scaling, and zero-downtime upgrades automated through Cluster API.
- Cut manual operational effort by automating about **80%** of data center provisioning and configuration processes.

Blockchain Developer

06/2023 – 09/2023

EsoiLabs | *Innovaz Market Factory – cross-chain NFT and token platform*

- Developed event-driven services (capture, process, persist) that move on-chain events into off-chain databases and expose them through APIs.
- Kept transaction handling consistent and data intact across **5+ blockchain networks** (Ethereum, BSC, Polygon, Aptos, Sui).
- Delivered **15+** reviewed smart contracts (Solidity, Move) for DeFi and NFT use cases.

SELECTED PROJECTS

DattingQ | Software Engineer

Dating platform

- Built a backend exposing **REST/JSON APIs** through gRPC-Gateway in front of gRPC services.
- Used **Kafka** for messaging and **Redis** and **MongoDB** for storage, sustaining up to **1.5K requests/second** at **80–100 ms**.
- Implemented multi-cluster failover and observability (OpenTelemetry, Prometheus, Grafana, ELK, ClickHouse) for **99.95% uptime**.

MeetQ | Software Engineer

AI-powered online meeting platform

- Built a real-time meeting backend on LiveKit integrating OpenAI models for transcription, voice translation, and summaries; designed the real-time translation pipeline and its context handling.

EDUCATION & CERTIFICATIONS

University of Science, Ho Chi Minh City (HCMUS)

B.Sc. Information Technology – Knowledge Engineering

DevOps on AWS Specialization | Coursera